

Vending Machine Mod

User Guide - Beta

For Minecraft NeoForge 1.21 / 1.21.1 test builds

This guide explains how to place, configure, stock, protect, move, and use the vending machine, including admin settings, cashbox behavior, signs, recipes, and troubleshooting.

Guide item	Details
Current phase	Beta / test build
Primary block	Vending Machine
Admin item	Admin Tool
Currency item	Vendor Coin, plus any configured item such as emeralds or diamonds
Main user roles	Owner/admin configures; customers buy from the public shop GUI

1. Mod Description

The Vending Machine Mod adds a player-owned vending machine that can sell configured items to other players. Each machine has configurable sale slots, internal stock, a customer output tray, and a cashbox that stores payment items. Machines remember their owner, settings, inventory, sign selection, and cashbox contents when moved.

- Player-owned vending machines with owner/admin protection.
- Nine configurable sale slots arranged as three shelves with three items per shelf.
- Customer shop GUI with Buy buttons and visible stock counts.
- Admin GUI for stock, output tray, cashbox, price configuration, and settings.
- Movable saved-data vending machines: break and replace without losing setup or contents.
- Glowing sign presets and displayed sale items rendered behind glass.
- Crafting recipes, recipe unlock advancements, creative tab support, and JEI visibility.

2. Items and Blocks

Name	Purpose
Vending Machine	The two-block-tall machine players place and configure.
Admin Tool	Used by the owner or server operator to open the admin/config GUIs.
Vendor Coin	Optional custom currency item that can be used as a configured price item.

3. Getting Started

1. Craft or obtain a Vending Machine and Admin Tool.
2. Place the Vending Machine with two blocks of vertical clearance. It places as a two-block-tall machine.
3. The player who places the machine becomes its owner.
4. Right-click the machine normally to open the customer shop GUI.
5. Use the Admin Tool on the machine to open the owner/admin GUI.

Important: A machine can only be configured, stocked, collected, or broken by its owner or by a server operator/admin. Other players can still buy from the public shop GUI.

4. Ownership and Protection

Action	Owner	Other player	Operator/admin
Open customer shop GUI	Allowed	Allowed	Allowed
Open admin/config GUI	Allowed	Blocked	Allowed
Break/move machine	Allowed	Blocked	Allowed
Collect cashbox	Allowed	Blocked	Allowed
Change prices/signs	Allowed	Blocked	Allowed

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Old/unowned machines: If a machine does not have owner data, the first player to use the admin tool on it can claim it. This supports older test machines from before ownership existed.

5. Admin GUI Overview

Use the Admin Tool on a machine you own to open the Vending Machine Admin screen. The admin screen includes:

- Configure Prices button - opens the sale/price setup screen.
- Gear/settings button - opens machine settings such as sign preset selection.
- Stock Inventory - items available for sale. External storage integration will fill this inventory later.
- Output Tray - purchased items waiting for the customer to take them.
- Cashbox - payment items collected from purchases.
- Player Inventory - your normal inventory for moving items in and out.

Stock inventory stack size: The internal stock/cashbox is designed for high stack limits, while the output tray should still output normal item stacks to customers.

6. Configure Prices

The Configure Prices screen controls the nine sale slots. Each sale slot has two ghost/template slots:

- Sell item - what the machine sells to the customer.
- Price item - what the customer must pay.

Sale slot	Shelf display position	Config slots
1	Top shelf - left	Sell item + price item
2	Top shelf - middle	Sell item + price item
3	Top shelf - right	Sell item + price item
4	Middle shelf - left	Sell item + price item
5	Middle shelf - middle	Sell item + price item
6	Middle shelf - right	Sell item + price item
7	Bottom shelf - left	Sell item + price item
8	Bottom shelf - middle	Sell item + price item
9	Bottom shelf - right	Sell item + price item

Quantity buttons adjust how many items are sold and how many payment items are required. The controls support:

- Left click + / - : adjust by 1.
- Shift + click + / - : adjust by 16.
- Ctrl + Shift + click + / - : adjust by 64.

Ghost slots: The configure screen stores templates only. Placing an item in a ghost slot does not consume the item from your inventory. The actual sale item must be placed into Stock Inventory for customers to buy it.

7. Customer Shop GUI

Any player can open the customer shop GUI by right-clicking the vending machine normally. The customer screen shows:

- Nine displayed sale slots.
- The configured sell item and price item.
- Stock count for each sale slot.
- Buy buttons for each configured sale slot.
- Output tray where purchased items appear.

Buy result	What happens
Successful purchase	Payment is removed from the customer, item is removed from stock, purchased item goes into the output tray, and payment goes into the cashbox.
Missing payment	Customer receives a message explaining what payment is needed.
Out of stock	Customer receives an out-of-stock message.
Output tray full	Purchase is blocked until the tray has room.
Cashbox full	Purchase is blocked until the owner collects payments.
Unconfigured sale slot	Purchase is blocked and a message is shown.

8. Cashbox and Payment Handling

Payments no longer disappear. When a customer buys an item, the payment item is inserted into the machine cashbox. The owner can open the admin GUI and remove payments from the cashbox.

- The cashbox is owner/admin access only.
- Customers cannot directly access the cashbox.
- If the cashbox is full, purchases are blocked to avoid deleting payment items.
- The cashbox contents are saved when the machine is moved.

9. Settings and Sign Presets

The gear button on the top-right of the admin GUI opens the Vending Machine Settings screen. The current beta settings screen allows the owner to cycle the display sign preset.

- Use < and > to cycle sign presets.
- Click Back to return to the admin GUI.
- The selected sign preset saves with the machine.
- The sign renders as a bright/glowing texture over the top black sign area on the model.

Sign preset	Suggested texture file
Ingots	textures/sign/vendor_sign_ingots.png
Food	textures/sign/vendor_sign_food.png
Tools	textures/sign/vendor_sign_tools.png
Ores	textures/sign/vendor_sign_ores.png
Blocks	textures/sign/vendor_sign_blocks.png
Magic	textures/sign/vendor_sign_magic.png
General	textures/sign/vendor_sign_general.png

10. Moving a Vending Machine

The vending machine can be moved without losing setup or inventory.

6. Make sure you are the machine owner or an operator/admin.
7. Break the vending machine in survival mode.
8. Pick up the dropped vending machine item.
9. Place it somewhere else.
10. Open the admin GUI to confirm the config, stock, output tray, cashbox, owner, and sign setting are still saved.

Note: If you want to retire a machine instead of moving it, manually empty the stock, output tray, and cashbox first, then break it.

11. Visual Display Features

The machine model can show the actual configured sale items behind the glass. The display uses three shelves with three items per shelf.

- Sale slots 1-3 appear on the top shelf.
- Sale slots 4-6 appear on the middle shelf.
- Sale slots 7-9 appear on the bottom shelf.
- Displayed items update when the sale configuration changes.
- The sign texture updates when the selected sign preset changes.

12. Crafting Recipes and JEI

The mod includes crafting recipes and recipe unlock advancements. Recipes should appear naturally after obtaining the relevant ingredients. They should also appear in JEI after the items are added to creative tabs.

Recipe	Unlock trigger	Notes
Vending Machine	Iron ingot	Main block. Can use tag-based ingredients later for modpack compatibility.
Admin Tool	Redstone	Owner/admin configuration tool.

Vendor Coin	Gold ingot	Optional custom currency item.
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For modpack-friendly recipes, ingredient tags can be used instead of specific vanilla items. Example:

"G": { "tag": "c:glass_blocks" }

"I": { "tag": "c:ingots/iron" }

"R": { "tag": "c:dusts/redstone" }

13. Troubleshooting

Problem	Likely cause / fix
Item/recipe does not show in JEI	Make sure the jar is the latest build and items are registered to creative tabs. Restart the game.
Recipe does not appear in vanilla recipe book	Get the unlock ingredient or run /recipe give @p * for testing.
Cannot open admin GUI	You are not the owner, or you are not an operator/admin.
Cannot break a machine	Break protection blocks non-owners. Owner or operator/admin can break it.
Purchase button does nothing	Check payment, stock, output tray space, and cashbox space. The GUI should show a feedback message.
Machine loses settings after moving	Make sure you broke the saved machine item in survival and placed that dropped item, not a new machine.
Displayed sale items are wrong/missing	Open Configure Prices and confirm the sell-item ghost slots are configured.
Sign does not change	Use the gear/settings screen and make sure the correct sign texture exists in textures/sign.

14. Server and Modpack Notes

- Every client and the server must use the same mod jar.
- Test builds should be used on backed-up worlds. Beta features may still change.
- Server operators can override ownership protections for maintenance.
- Future storage integration will likely begin with generic item-handler/hopper-style insertion, then expand to specific storage mods.

15. Beta Roadmap

Version	Goal
0.1.0-beta	Current playable beta: vending machine, owner protection, GUIs, cashbox, movable saved machine, signs, visual sale items, recipes/JEI.

0.2.0-beta	Storage integration / automatic filling support.
0.3.0-beta	Buyback/vendor machine for players selling items to a machine.
0.4.0-beta	Settings polish, sign options, config options, multiplayer refinements.
1.0.0	First production/stable release after initial planned features are complete.